

Exercise 1 - Caesar Cipher

1. Identify the cipher

Caesar Cipher (shift = 3)

2. Decrypt the ciphertext

Ciphertext:

FRPSXWHUVHFXULWB

Now shift back 3 letters

CT	F R P S X W H U V H F X U L W B
PT	C O M P U T E R S E C U R I T Y

now PT = computer security

Exercise 2 - Monalphabetic Substitution Cipher

PT	A B C D E F G H I J K L
CT	Q W E R T Y U I O P A S

PT	M N O P Q R S T U V W X
CT	D F G H J K L Z X C V B

PT	Y Z
CT	N M

CT: - L T E X K O Z N

Find each cipher letter in cipher row & plain row

CT	L T E X K O Z N
PT	S E C U R I T Y

PT = security

Exercise 2 - Vigenere Cipher

Ciphertext:

LXFOPVFRNHR

Key:

LEMON

Repeat the key:

Cipher	L	X	F	O	P	V	E	F	R	N	H	R
Key	L	E	M	O	N	L	E	M	O	N	L	E
Plain	A	T	T	A	C	K	A	T	D	A	W	N

PT:- ATTACK AT DAWN

Exercise 3 - Rail fence cipher 3 rails

CT:- WE CRLTEERD SOEEF EAO CATVDEN

Rail 1 W E C R L T E

Rail 2 E R D S O E E F E A O E

Rail 3 C A T I V D N

Plain text

WE ARE DISCOVERED FLEE
AT ONCE

Exercise 4 - Columnar Transposition

Cipher:-

key: HACK

Arrange alphabetically:

H	A	C	K
3	1	2	4

Recovered messages:-

ATTACK AT DAWN

Plaintext:-

ATTACK AT DAWN

Exercise - 6 - Hill Cipher

key matrix:-

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

Cipher text:-

HTIOZ

Split into pairs:-

HI OZ

Convert letters to numbers

$$H = 7$$

$$I = 8$$

$$O = 4$$

$$Z = 25$$

inverse matrix:-

$$K^{-1} = \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix}$$

multiply :-

for HI:

$$\begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \times \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$$5 = F, 10 = K$$

for OZ:

$$\begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \times \begin{bmatrix} 14 \\ 25 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

$$5 = F, 3 = D$$

PT:- FKFD